

BIOXITIDE 25/125 mcg/dose HFA INHALATION AEROSOL
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Salmeterol (as Salmeterol Xinafoate)/ Fluticasone propionate (25/125, 25/250 micrograms)
Description: Bioxotide[®] HFA Inhalation Aerosol, 120 doses is a homogenous suspension aerosol for inhalation, supplied in a pressurised container. On visual examination, there must be no sign of physical damage or leakage.

Pharmacodynamics: Bioxotide[®] contains Salmeterol and Fluticasone Propionate, which have different modes of action. Salmeterol protects against symptoms, Fluticasone Propionate improves lung function and prevents exacerbations of the condition. Bioxotide[®] can offer a more convenient regime for patients on concurrent beta-agonist and inhaled corticosteroid therapy. The respective mechanisms of action of both drugs are discussed below:
Salmeterol: Salmeterol is a selective long-acting (12hours) beta-2-adrenoreceptor agonist with alongside chain which binds to the exo-site of the receptor.
Fluticasone propionate: Fluticasone propionate given by inhalation at recommended doses has a potent glucocorticoid anti-inflammatory action within the lungs, resulting in reduced symptoms and exacerbations of asthma, without the adverse effects observed when corticosteroids are administered systemically.

Pharmacokinetics: When Salmeterol and Fluticasone propionate were administered in combination by the inhaled route, the pharmacokinetics of each component were similar to those observed when the drugs were administered separately. For pharmacokinetic purposes therefore each component can be considered separately.
Salmeterol: Salmeterol acts locally in the lung therefore plasma levels are not an indication of therapeutic effects. In addition there are only limited data available on the pharmacokinetics of salmeterol because of the technical difficulty of assaying the drug in plasma due to the low plasma concentrations at therapeutic doses (approximately 200 picogram/ml or less) achieved after inhaled dosing.
Fluticasone propionate: The absolute bioavailability of a single dose of inhaled fluticasone propionate in healthy subjects varies between approximately 5-11% of the nominal dose depending on the inhalation device used. In patients with asthma a lesser degree of systemic exposure to inhaled fluticasone propionate has been observed. Systemic absorption occurs mainly through the lungs and is initially rapid than prolonged. The remainder of the inhaled dose may be swallowed but contributes minimally to systemic exposure due to the low aqueous solubility and pre-systemic metabolism, resulting in oral availability of less than 1%. There is a linear increase in systemic exposure with increasing inhaled dose. The disposition of fluticasone propionate is characterised by high plasma clearance (1150ml/min), a large volume of distribution at steady-state (approximately 300l) and a terminal half-life of approximately 8 hours. Plasma protein binding is 91%.
Fluticasone propionate is cleared very rapidly from the systemic circulation. The main pathway is metabolism to an inactive carboxylic acid metabolite, by the cytochrome P450 enzyme CYP3A4. Other unidentified metabolites are also found in the faeces. The renal clearance of fluticasone propionate is negligible. Less than 5% of the dose is excreted in urine, mainly as metabolites. The main part of the dose is excreted in faeces as metabolites and unchanged drug.

Indication:
Asthma (Reversible Obstructive Airways Disease)
Bioxotide[®] is indicated in the regular treatment of reversible obstructive airways disease (ROAD), including asthma in children and adults, where use of a combination (bronchodilator and inhaled corticosteroid) is appropriate. This may include:
- Patients on effective maintenance doses of long-acting beta-agonists and inhaled corticosteroids.
- Patients who are symptomatic on current inhaled corticosteroid therapy.
- Patients on regular bronchodilator therapy who require inhaled corticosteroids.
Chronic Obstructive Pulmonary Disease (COPD)
Bioxotide[®] is indicated for the regular treatment of chronic obstructive pulmonary disease (COPD) including chronic bronchitis and emphysema.

Recommended Dose:
Bioxotide[®] is for inhalation only. Patients should be made aware that Bioxotide[®] must be used regularly for optimum benefit, even when asymptomatic. Patients should be regularly reassessed by a doctor, so that the strength of Bioxotide[®] they are receiving remains optimal and is only changed on medical advice.
Asthma (Reversible Obstructive Airways Disease): The dose should be titrated to the lowest dose at which effective control of symptoms is maintained. Where the control of symptoms is maintained with twice daily Bioxotide[®], titration to the lowest effective dose could include Bioxotide[®] given once daily. Patients should be given the strength of Bioxotide[®] containing the appropriate fluticasone propionate dosage for the severity of their disease. If a patient is inadequately controlled on inhaled corticosteroid therapy alone, substitution with Bioxotide[®] at a therapeutically equivalent corticosteroid dose may result in an improvement in asthma control. For patients whose asthma control is acceptable on inhaled corticosteroid therapy alone, substitution with Bioxotide[®] may permit a reduction in corticosteroid dose while maintaining asthma control. Recommended doses:
Adults and adolescents 12 years and older:-
Two inhalations of 25 micrograms salmeterol and 125 micrograms fluticasone propionate twice daily. Or Two inhalations of 25 micrograms salmeterol and 250 micrograms fluticasone propionate twice daily.
Children 4 years and older:-
The available strengths are not suitable for children in this age group. There are no data available for use in children aged less than 4 years.
Chronic Obstructive Pulmonary Disease (COPD): For adult patients the recommended dose is two inhalations 25/125 micrograms to 25/250 micrograms salmeterol/fluticasone propionate twice daily.
Special patient groups:-There is no need to adjust the dose in elderly patients or in those with renal or hepatic impairment.

Route of Administration: Respiratory (Inhalation)
Contraindication: Bioxotide[®] is contraindicated in patients with hypersensitivity to any of the active substances or to the excipient.

Warnings: The management of Road should normally follow a stepwise programme and patient response should be monitored clinically and by lung function tests. Bioxotide[®] is not for relief of acute symptoms for which a fast-acting short-acting bronchodilator (eg. Salbutamol) is required. Patients should be advised to have their relief medication available at all times. Increasing use of short-acting bronchodilators to relieve symptoms indicates deterioration of control and patients should be reviewed by physician. Sudden and progressive deterioration in control of asthma is potentially life-threatening and the patient should be reviewed by a physician. Consideration should be given to increasing corticosteroid therapy. Also, where the current dosage of Bioxotide[®] has failed to give adequate control of ROAD, the patient should be reviewed by a physician. Bioxotide[®] should not be initiated in patients with unstable and acutely deteriorating asthma, which may be a life-threatening condition. Bioxotide[®] should not be used to transfer patients from oral to inhaled steroids. For patients with asthma of COPD, consideration should be given to additional corticosteroid therapies and administration of antibiotics if an exacerbation is associated with infection. Treatment with Bioxotide[®] should not be stopped abruptly in patients with asthma due to risk of exacerbation, therapy should be titrated-down under physician's supervision. For patients with COPD, cessation of therapy may be associated with symptomatic decompensation and should be supervised by a physician. As with all inhaled medication containing corticosteroids, Bioxotide[®] should be administered with caution in patients with active or quiescent pulmonary tuberculosis. Bioxotide[®] should be administered with caution in patients with thyrotoxicosis. Cardiovascular effects eg. Increases in systolic blood pressure and heart rate, may occasionally be seen with all sympathomimetic drugs, especially at higher than therapeutic doses. For this reason, Bioxotide[®] should be used with caution in patients with pre-existing cardiovascular disease. A transient decrease in serum potassium may occur with all sympathomimetic drugs at higher therapeutic doses. Therefore, Bioxotide[®] should be used with caution in patients predisposed to low levels of serum potassium. Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods; these effects are much less likely to occur than with oral corticosteroids (see Overdosage). Possible systemic effects include Cushing's syndrome, Cushingoid features, and adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma. It is important, therefore for ROAD patients, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control is maintained. The possibility of impaired adrenal response should always be borne in mind in emergency and elective situations likely to produce stress and appropriate corticosteroid treatment considered (see Overdosage). Certain individuals can show greater susceptibility to the effects of inhaled corticosteroid than do most patients. Because of the possibility of impaired adrenal response, patients transferring from oral steroid therapy to inhaled fluticasone propionate therapy should be treated with special care and adrenocortical function regularly monitored. Following introduction of inhaled fluticasone propionate, withdrawal of systemic therapy should be gradual and patients encouraged to carry a steroid warning card indicating the possible need for additional therapy in time of stress. Patients in a medical or surgical emergency, who in the past have required high doses of inhaled steroids and/or intermittent treatment with oral steroids, remain at risk of impaired adrenal reserve for a considerable time. The extent of the adrenal impairment may require specialist's advice before elective procedures. The possibility of residual impaired adrenal response should always be borne in mind in emergency and elective situation likely to produce stress and appropriate corticosteroid treatment must be considered. Keep out of reach of children. Jauhi ubat daripada capaian kanak-kanak.

Pneumonia in patients with COPD
An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies. There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products. Physicians should remain vigilant for the possible development of pneumonia in patient with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations. Risk factors for pneumonia in patients with COPD include current smoking status, older age, low body mass index (BMI) and severe COPD.

Side effects: As Bioxotide[®] contains salmeterol and fluticasone propionate, the type and severity of adverse reactions associated with each of the compounds may be expected. Adverse events which have been associated with salmeterol/fluticasone propionate are given below:

System Organ Class	Adverse Event
Infections & Infestations	Candidiasis of the mouth and throat Pneumonia Bronchitis Oesophageal candidiasis
Immune System Disorders	Hypersensitivity reactions with the following manifestations: Cutaneous hypersensitivity reactions Angioedema (mainly facial and oropharyngeal oedema) Respiratory symptoms (dyspnoea) Respiratory symptoms (bronchospasm) Anaphylactic reactions including anaphylactic shock
Endocrine Disorders	Cushing's syndrome, Cushingoid features, Adrenal suppression, Growth retardation in children and adolescents, Decreased bone mineral density

Metabolism & Nutrition Disorders	Hypokalaemia Hyperglycaemia
Psychiatric Disorders	Anxiety Sleep disorders Behavioural changes, including psychomotor hyperactivity and irritability (predominantly in children) Depression, aggression (predominantly in children)
Nervous System Disorders	Headache Tremor
Eye disorder	Cataract Glaucoma
Cardiac Disorders	Palpitations Tachycardia Cardiac arrhythmias (including supraventricular tachycardia and extrasystoles). Atrial fibrillation Angina pectoris
Respiratory, Thoracic & Mediastinal Disorders	Nasopharyngitis Throat irritation Hoarseness/dysphonia Sinusitis Paradoxical bronchospasm
Skin and subcutaneous tissue disorders	Contusions
Musculoskeletal & Connective Tissue Disorders	Muscle cramps Traumatic fractures Arthralgia Myalgia

Both hoarseness and incidence of candidiasis may be relieved by gargling with water after use of Salmeterol/ Fluticasone Propionate Inhaler. "Pneumonia (in COPD patients)" to be listed as "Common" adverse drug reaction in the "Infections and Infestations" SOC.

Interactions with other medications: Both nonselective and selective beta-blockers should be avoided in patients with asthma, unless there are compelling reasons for their use. Due to the very low plasma concentrations achieved after inhaled dosing clinically significant drug interactions are unlikely. Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side effects.

Pregnancy and lactation: Administration of drugs during pregnancy and lactation should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus or child. There is insufficient experience of the use of Salmeterol Xinafoate and Fluticasone propionate in human pregnancy and lactation. There are no data available for human breastmilk.

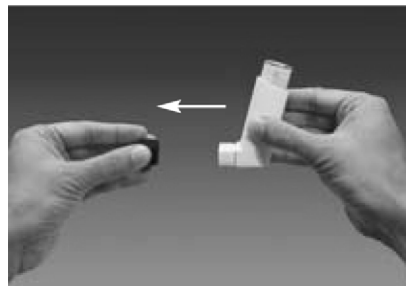
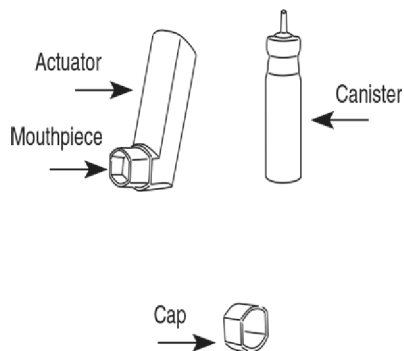
Symptoms and Treatment of Overdose: The expected sign and symptoms of salmeterol overdose are those typical of excessive β2-adrenergic stimulation, including tremor, headache, tachycardia, increases in systolic blood pressure and hypokalaemia. The preferred antidotes are cardioselective beta-blocking agents, which should be used with caution in patients with a history of bronchospasm. If Bioxotide[®] therapy has to be withdrawn due to overdose of beta-agonist component of the drug, provision of appropriate replacement corticosteroid therapy should be considered. Acute inhalation of fluticasone propionate doses in excess of those approved may lead to temporary suppression of the hypothalamic-pituitary-adrenal axis. This does not usually require emergency action as normal adrenal function typically recovers within a few days. If higher than approved doses of Bioxotide[®] are continued over prolonged periods, significant adrenocortical suppression is possible. It is not recommended that patients receive higher than approved doses of Bioxotide[®]. It is important to review therapy regularly and titrate down to the lowest approved dose at which effective control of disease is maintained. (See dosage & administration)

Storage Condition: Avoid storage in direct sunlight or heat. Store below 30°C.
As with most inhaled medications in aerosol canisters, the therapeutic effect of this medication may decrease when the canister is cold. The canister should not be broken, punctured or burnt, even when apparently empty.
Shelf life: 2 years

Dosage form: Preparation for Inhalation in form of Pressurised Metered-Dose Inhaler
Packaging: Each canister contains 120 metered doses. The aluminium canister fitted with valve & labelled, and then again fitted with an actuator and dust cap. This is then packed inside an inner carton with package insert inside. It does not contain CFC as propellant.

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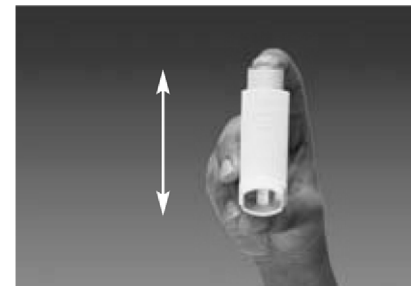
HOW TO USE YOUR INHALER CORRECTLY



1. Remove the cap from the mouthpiece of the actuator.



2. Make sure the mouthpiece is clean inside and outside.



3. Hold the inhaler by placing your index finger on top of the metal canister and thumb on the bottom of the plastic mouthpiece. Shake it well.



4. Raise the Inhaler to your mouth. Put the mouthpiece between your teeth, but do not bite it. Close your lips around the mouthpiece. Breathe out slowly and gently through the Inhaler until your lungs feel comfortably empty.



5. Tilt your head back slightly. Start to breathe in slowly through your mouth. As you start to breathe in, press down firmly on the top of the can to release your medicine continue to breathe in steadily and deeply.



6. Hold your breath. Remove the inhaler from your mouth. Continue to hold your breath as long as possible, up to 10 seconds. Then breathe out gently. If you are taking a second puff, wait about one minute, then repeat steps 3 to 6.



7. Replace the mouthpiece cap after each use.

A handy tip for Children

Children and others who have weaker hands may have difficulty pressing down on the top of the can with just one hand. They can use both hands to make their Inhaler work.



Cleaning your Inhaler

Keeping the plastic actuator clean is very important to prevent medicine buildup and blockage. The actuator should be washed, shaken to remove excess water and air-dried thoroughly at least once a week. The inhaler may stop spraying if not properly cleaned.

Shake well the inhaler before each use

How to clean your Inhaler?

1. Remove the metal canister from the plastic casing of the inhaler and remove the mouthpiece cover.
2. Rinse the actuator thoroughly with warm water.
3. Dry the actuator thoroughly inside and outside.
4. Replace the metal canister and the mouthpiece cover.
5. Do not put the metal canister in water.

Your Inhaler should be cleaned at least once a week

Precaution

1. Pressurised canister, do not puncture, break or incinerate even when apparently empty.
2. Avoid storage in direct sunlight or heat.
3. Store below 30° c
4. Keep away from eyes
5. Keep away from children
6. Handle the inhaler with care when using with other device i.e spacer or nebulizer to avoid damage to the inhaler.